BIOT6900 Assignment 2 Review

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I choose to cover colon adenocarcinoma. This was a disease which I found to have similar quantities of data to the breast cancer example we used in the first part of the assignment. Colon cancer is the third most common cancer diagnosed in both men and women in the United States and the second most common cause of cancer deaths.[1] As such, colon cancer is a well studied disease with a large quantity of written literature, which makes it an ideal candidate for using multi-modal data to identify target genes of interest.

For datasets, I choose the 'CPTAC - Colon Adenocarcinoma' dataset from Linkedomics[2]. This was an open dataset. From this dataset, I used three different categories of data: RNA, proteome, and mutation. Similar to the breast cancer data, this gives three different potentially correlated areas to investigate and draw conclusions from. Additionally, I used the TCGA colon adenocarcinoma (COAD) gene expression (RNA) data from USCS[3]. Unlike the CPTAC RNA data which only accounts for tumors, this dataset includes tumors mixed with normal tissue. This offers a different mode of comparison, allowing for identification of which genes are expressed disproportionately in cancer tissue rather than healthy tissue. By including the TCGA data, we can identify genes whose incidence may be more revealing than a simple quantitative count. All four sources of data were processed and joined, giving an end total of 4470 genes with data across all 4 modes. One critical note is that the CPTAC data is from different patients then the TCGA data, meaning that we cannot draw direct patient level conclusions, only broad scale correlational observations.

It is also worth explaining how the layers line up. The CPTAC RNA, proteome, and mutation data all come from the same group of patients, so those three are matched. The CPTAC proteome data was already given as a tumor versus normal fold change, since that study collected matched normal tissue from each patient alongside the tumor.[5] The TCGA RNA data is the only unmatched layer, so it was combined with the CPTAC data at the gene level instead of the patient level. Each layer was reduced to one number per gene before scoring. The CPTAC RNA data was still kept for calculating the RNA to protein correlation, since that calculation needs the same patients measured both ways.

For the code, I was able to reuse a good chunk of the code from the first part of the assignment. Repurposing many of the helper functions while making changes to accommodate the additional source of data. I used the same simple scoring algorithm as the first assignment. The scoring works by converting each layer into a percentile from 0 to 1, then combining them with equal weights. This means the raw scale of each layer does not matter.

I created a list of the top 15 identified targets via the 4 data sources and the simple scoring algorithm:

	transcriptomic	proteomic	genomic	rna_prot_corr	score
FCGBP	4.449	1.82	0.151	0.543	0.992
MYH11	4.422	2.762	0.113	0.427	0.992
ANK2	3.499	1.521	0.142	0.393	0.986
FLNC	2.444	2.17	0.132	0.164	0.984
CP	3.924	1.175	0.113	-0.025	0.977
KCNMA1	3.546	1.028	0.16		0.976
MYLK	2.896	2.262	0.085	0.459	0.975
DMD	1.912	1.569	0.179	0.404	0.975
COL1A1	2.276	1.887	0.085	0.277	0.968
COLEC12	2.573	1.867	0.075		0.965
EPB41L3	3.52	1.376	0.075	0.628	0.964
MUC2	3.131	1.023	0.094	0.587	0.963
SORBS2	1.489	1.935	0.113	0.143	0.962
PPP1R12B	2.109	1.996	0.075	0.386	0.961
TNS1	2.28	1.629	0.075	0.316	0.96

There are numerous genes which were scored highly by the simple scoring method. Doing some quick verification on Open Targets, the top scorers all have evidence linking them to colon cancer, validating the results. For example, FCGBP has been identified by numerous studies as having an association with colon cancer[4]. It should be noted that the difference between the top scoring genes is small and the scoring algorithm is intentionally simplistic, so the top genes should not be ranked but rather thought of as a group of potentially interesting targets.

Most of the top 15 fall into two groups. The first is goblet cell and mucus genes, FCGBP and MUC2. Low FCGBP levels have been found in colorectal adenoma and carcinoma[4], so seeing these genes is a good sign that the pipeline is picking up real colon cancer biology. The second group is smooth muscle genes: MYH11, MYLK, FLNC, KCNMA1, SORBS2, PPP1R12B, DMD, TNS1, and ANK2. These are likely an artifact of the data, showing up because the normal tissue samples contain more of the

muscle layer of the colon wall than the tumor samples do. Differences in tumor purity like this are a known confounder in TCGA transcriptomic data.

EPB41L3 is an interesting hit. Unlike most of the other top genes, it is not a smooth muscle or mucus gene. It had large changes in both RNA (3.52) and protein (1.376), and it had the highest RNA to protein correlation in the top 15 at 0.628. This means that patients with more EPB41L3 RNA also tended to have a bigger change in protein, which suggests a real change rather than noise. EPB41L3 has been studied as a possible tumor suppressor in other cancers, but I wasn't able to find much about it in relation to colon cancer, which makes it worth a second look.

CP (ceruloplasmin) is the most discordant gene in the top 15. It had a large RNA change (3.924) and a moderate protein change (1.175), but an RNA to protein correlation of essentially zero (-0.025).[6] CP is mostly made by the liver and circulates in the blood, so a lot of the CP protein measured in a tissue sample could come from blood in the tissue rather than from the colon cells themselves. There are also technical reasons. The correlation here compares tumor RNA levels to protein fold change, which are not the same kind of measurement.

With this setup, there are some things I can claim and some I cannot. I can say that these genes show large changes between colon tumor and normal tissue at the gene level (per the TCGA data) and that the RNA and protein changes mostly agree even though they come from two different cohorts, which is a good sign that we are identifying real signals and not just noise, since two independent datasets are pointing at the same genes. As previously stated, I cannot claim that RNA and protein change together in the same patients, since the TCGA and CPTAC patients are different people. It is also crucial to highlight that the identification of the genes is not a claim that any of these genes cause cancer, since a change in expression is not evidence of being a driver (for example, the smooth muscle genes identified are likely a result of their incidence in the muscle layer of colon, not because of any existing cancer). Finally, a high score does not mean a gene is a good drug target. More investigation is necessary to determine by what mechanism the gene is expressed and whether or not it is actually worth a further look.

Citations

[1] American Cancer Society. "Colorectal Cancer Statistics." *American Cancer Society*, 14 Jan. 2026, www.cancer.org/cancer/types/colon-rectal-cancer/about/key-statistics.html.

[2] https://www.linkedomics.org/data_download/CPTAC-COAD/

[3] https://xenabrowser.net/datapages/?cohort=TCGA%20Colon%20Cancer

[4] Yuan Z, Zhao Z, Hu H, et al. IgG Fc Binding Protein FCGBP is Down-Regulated in Metastatic Lesions and Predicts Survival in Metastatic Colorectal Cancer Patients. Oncotargets and Therapy. 2021 ;14:967-977. DOI: 10.2147/ott.s285171. PMID: 33603401; PMCID: PMC7886382.

[5] Vasaikar S, Huang C, Wang X, et al. Proteogenomic Analysis of Human Colon Cancer Reveals New Therapeutic Opportunities. Cell. 2019;177(4):1035-1049.e19. DOI: 10.1016/j.cell.2019.03.030.

[6] Liu Y, Beyer A, Aebersold R. On the Dependency of Cellular Protein Levels on mRNA Abundance. Cell. 2016;165(3):535-550. DOI: 10.1016/j.cell.2016.03.014.